

LAB04_marreiro.qmd

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Laboratório 4: Importação de Dados Retangulares - Excel

```
# Q1. Carregue os pacotes readxl e tidyverse.  
library(readxl)  
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --  
v dplyr      1.2.1      v readr      2.2.0  
v forcats    1.0.1      v stringr    1.6.0  
v ggplot2    4.0.3      v tibble     3.3.1  
v lubridate  1.9.5      v tidyr      1.3.2  
v purrr      1.2.2  
-- Conflicts ----- tidyverse_conflicts() --  
x dplyr::filter() masks stats::filter()  
x dplyr::lag()     masks stats::lag()  
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(dplyr)
```

```
# Q2. Crie uma variável chamada fname que contenha o caminho para o arquivo WDIEXCEL.xlsx  
fname <- "/Users/fernandamarreiro/Desktop/WDIEXCEL.xlsx"  
file.exists(fname)
```

```
[1] TRUE
```

```
# Q3. Ler a planilha "Country"  
countries <- read_excel(fname, sheet = "Country")
```

```
# Q4. Filtrar Brasil e mais 4 países de sua escolha
países_selecionados <- c("BRA", "ARG", "CHL", "COL", "URY")

countries <- countries %>%
  filter(`Country Code` %in% países_selecionados)
```

```
# Q5. Ler a planilha "Data"
WDI <- read_excel(fname, sheet = "Data")
```

```
# Q6. Selecionar Country Code, Indicator Code e anos de 1960 a 2018
WDI <- WDI %>%
  select(`Country Code`, `Indicator Code`, `1960`:`2018`)
```

```
# Q7. Pivotar a tabela para o formato Tidy (coluna YEAR)
WDI <- WDI %>%
  pivot_longer(
    cols = `1960`:`2018`,
    names_to = "YEAR",
    values_to = "Value"
  )
```

```
# Q8. Converter a coluna YEAR para inteiro
WDI <- WDI %>%
  mutate(YEAR = as.integer(YEAR))
```

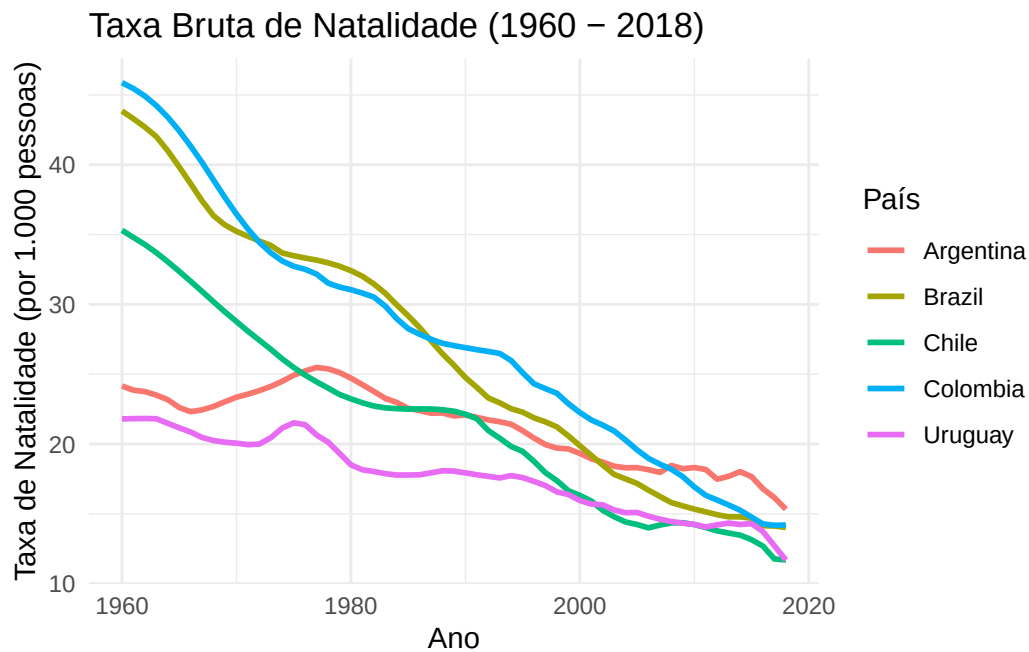
```
# Q9. Filtrar os países selecionados, buscar o nome (Short Name) e o indicador
WDI <- WDI %>%
  semi_join(countries, by = "Country Code") %>%
  left_join(countries %>% select(`Country Code`, `Short Name`), by = "Country Code") %>%
  filter(`Indicator Code` == "SP.DYN.CBRT.IN") %>%
  rename(SP.DYN.CBRT.IN = Value) %>%
  select(`Country Code`, `Short Name`, YEAR, SP.DYN.CBRT.IN)
```

```
# Q10. Ordenar por país e ano
WDI <- WDI %>%
  arrange(`Country Code`, YEAR)
```

```
# Q11. Gerar o gráfico de linhas com ggplot2
ggplot(WDI, aes(x = YEAR, y = SP.DYN.CBRT.IN, color = `Short Name`)) +
  geom_line(size = 1) +
```

```
labs(
  title = "Taxa Bruta de Natalidade (1960 - 2018)",
  x = "Ano",
  y = "Taxa de Natalidade (por 1.000 pessoas)",
  color = "País"
) +
theme_minimal()
```

Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
i Please use `linewidth` instead.



```
Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()
```

[1] "2026-09-09 19:20:51 -03"